

Object Oriented Programming with Java

25.3 Faculty of Computing



Take Home Activity 04

Answer all questions.

- 1) Create a Java class called **Car**
 - a. Inside the class, declare three instance variables with suitable data types: **year**, **brand** and **price**.
 - b. Create a non-parameterized constructor that sets default values: **year-2000**, **brand-“Unknown”** and **price – 0.00**.
 - c. Create a method called **displayInfo** that takes no parameters and has no return type. It should print the values of all three variables.
 - d. In the main method, create an **object** of the **Car** class and **call displayInfo method**.
- 2) Create a Java class called **Student**.
 - a. Declare four instance variables with suitable data types: **studentId**, **name**, **age**, **gpa**.
 - b. Create a parameterized constructor that accepts all four values and assigns them to the instance variables.
 - c. Create a method called **introduce** that takes no parameters and has no return type. It should print a sentence introducing the student using all four variable values.
 - d. Create a method called **updateGpa** that takes one parameter called **newGpa** and has no return type. It should update the **gpa** variable with the new value and print a confirmation message.
 - e. In the main method, create **two Student objects** using the parameterized constructor, call **introduce** on both, then call **updateGpa** on one of them.
- 3) Create a Java class called **Product**.
 - a. Declare three instance variables with suitable data types: **productId**, **productName** and **unitPrice**.
 - b. Create both a non-parameterized constructor with suitable default values and a parameterized constructor that accepts all three values.
 - c. Create a method called **applyDiscount** that takes one parameter called **percentage** and has no return type. It should reduce the **unitPrice** by the given percentage and print the updated price.
 - d. Create a method called **getProductLabel** that takes no parameters and returns a **String**. It should return a label in the format: **productId - productName - Rs. unitPrice**.
 - e. In the main method, create one object using each constructor, apply a 15% discount on one, and print the product label for both.

- 4) Create a Java class called **Employee**.
- Declare four instance variables with suitable data types: **empId**, **empName**, **department**, and **basicSalary**.
 - Create a parameterized constructor that accepts all four values and assigns them to the instance variables.
 - Create a method called **calculateNetSalary** that takes one parameter called **bonusPercentage** and returns a **double**. It should calculate and return the net salary as **basicSalary + (basicSalary × bonusPercentage / 100)**.
 - Create a method called **getEmployeeSummary** that takes no parameters and returns a **String** containing the employee's ID, name, and department.
 - Create a method called **promote** that takes one parameter called **newDepartment** and has no return type. It should update the department variable and print "[empName] has been promoted to [newDepartment]."
 - In the main method, create two **Employee** objects, print their summaries, calculate the net salary with a **20%** bonus, and promote one employee to a new department.
- 5) Create a Java class called **BankAccount**.
- Declare three instance variables with suitable data types: **accountNo**, **holderName**, and **balance**.
 - Create a non-parameterized constructor that sets suitable default values for all three variables.
 - Create a parameterized constructor that accepts **accountNo**, **holderName**, and **balance** and assigns them to the instance variables.
 - Create a method called **deposit** that takes one parameter called **amount** and has no return type. It should add the amount to the current balance and print the updated balance.
 - Create a method called **withdraw** that takes one parameter called **amount** and returns a **String**. If the balance is sufficient, deduct the amount and return "*Withdrawal successful. Remaining balance: [balance]*". Otherwise return "*Insufficient funds.*"
 - Create a method called **getBalance** that takes no parameters and returns the current balance as a double.
 - Create a method called **getAccountInfo** that takes one parameter called branch and returns a **String** containing the account number, holder name, and branch.
 - In the main method, create two **BankAccount** objects using each constructor and demonstrate all four methods on both objects.